Circuit investigation

antocciphysics | TEACHER KEY | CPA

Learning target: Use measurements to test a circuit model safely.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 7; 3 course days later.*
12 V; drops add to the supply voltage.

02. Explain a worked example (worked)
Straight line slope 10 ohms; behavior beyond tested conditions remains untested.

03. Look and explain (model)
Ammeter in series; voltmeter parallel across resistor.

04. Build the solution (practice)
.50 A predicted. Actual values must be measured, not copied from prediction.

05. Use the idea (reason)
Check connections, meter range/placement, units, component values, supply voltage; retain raw evidence.

06. Choose a method (mixed) *Review: Unit 6, Day 16; 12 course days later.*
It becomes 1/4 as large by Wien’s law.

07. Exit check (exit)
Teacher verifies connections and safety before power; student records actual feedback.

08. STRETCH: Apply it another way (stretch)
Expected 0.8 A; not a fit. Check heating, contacts, meters, component identity and uncertainty.

Teaching note: Lab worksheet. Follow classroom equipment procedures; no mains electricity. Supplied data are practice values, not student measurements.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 7; 3 course days later.*
New resistance (3+2) R=5 R; current ratio 3/5≈0.6.

02. Explain a worked example (worked)
Straight line slope 10 ohms; behavior beyond tested conditions remains untested.

03. Represent and interpret (model)
Ammeter in series; voltmeter parallel across resistor.

04. Solve and justify (practice)
12.61 Ω; 5.07% above nominal. Meter uncertainty, resistor tolerance or heating may contribute.

05. Analyze the reasoning (reason)
Check connections, meter range/placement, units, component values, supply voltage; retain raw evidence.

06. Choose a method (mixed) *Review: Unit 6, Day 16; 12 course days later.*
T≈5000 K (about 5000 K); peak wavelength is rounded.

07. Exit check (exit)
Vary safe supply voltage; measure current and resistor voltage; control resistor/temperature; constant V/I or linear V-I through origin.

08. STRETCH: Apply it another way (stretch)
Central 10 ohms; minimum 2.9/0.32≈9.06, maximum 3.1/0.28≈11.07 ohms.

Teaching note: Lab worksheet. Follow classroom equipment procedures; no mains electricity. Supplied data are practice values, not student measurements.